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Wave packets bouncing off walls

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The solution of Schrödinger's equation that describes the time evolution of a wave packet bouncing off a completely repulsive barrier (the "wall") is so simple that it can be introduced very early in an introductory course in quantum mechanics. It provides a nice example of Schrödinger's equation at work and a striking illustration of interference when the reflected part of the packet interferes with the incident part to produce rapid oscillations in the probability density near the wall.

The need for this solution arises, for example, in relating the classical and quantal descriptions of a particle in a box. If the particle is represented by a wave packet, then Ehrenfest's theorem shows that its mean position will move with con-

stant speed as long as it is clear of the walls. One is then confronted with analyzing what happens when the packet hits a wall of the box.

In his recent textbook,¹ Robinett shows numerically calcu-

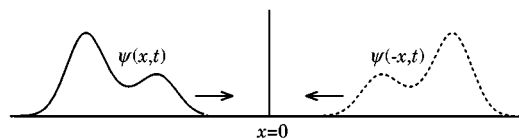


Fig. 1. A wave packet $\psi(x,t)$ approaching a wall at $x=0$, and its "image" $\psi(-x,t)$, represented by the dashed curve.

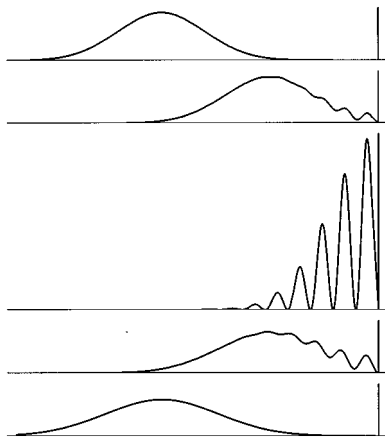


Fig. 2. The evolution of a free Gaussian wave packet bouncing off a wall. The probability density is shown for five equally spaced times. The initial wave packet has the form $\exp[ikx - x^2/a^2]$ with $ka = 12$.

lated results for a Gaussian packet bouncing off a wall. His graphs (similar to those in Fig. 2) show oscillations appearing in the incident wave packet when it reaches the wall; but after a time the wave packet, again apparently of Gaussian shape, emerges traveling away from the wall. It will now be shown that there is a simple analytic solution that explains this behavior and it applies to wave packets of any shape.

Consider a free particle hitting a wall. Suppose that at time $t=0$ the wave packet $\psi(x,t)$ lies entirely to the left of the wall, which is taken to be at $x=0$, as shown in Fig. 1. Thus $\psi(x,t)$ is a solution of the free Schrödinger equation (i.e., with no potential) such that $\psi(x,0)=0$ for all $x \geq 0$. The time evolution of this initial wave function $\psi(x,0)$ in the presence of the wall will then be that solution of the free Schrödinger equation which is always zero at the wall. The required solution is

$$\Psi(x,t) \equiv \psi(x,t) - \psi(-x,t) \quad \text{for } x < 0, \quad (1)$$

because (a) the flipped wave packet $\psi(-x,t)$ also satisfies the free Schrödinger equation, since this equation is unchanged under $x \rightarrow -x$, (b) at $t=0$, $\Psi(x,t) = \psi(x,t)$, as required, since $\psi(-x,0)$ is zero to the left of the wall, and (c) clearly $\Psi(0,t)=0$ for all t .

Since $\Psi(x,t)$ for $x < 0$ is the exact evolution in the presence of the wall it must remain normalized, i.e., $\int_{-\infty}^0 |\Psi|^2 dx$ must be constant. This also follows from the fact that $\Psi(x,t)$ is a solution of the free Schrödinger equation. Therefore, $\int_{-\infty}^{\infty} |\Psi|^2 dx$ will be constant (even though Ψ has violent oscillations when its two component packets overlap) and the antisymmetry of Ψ means that there will be equal contributions to the integral from each side of the wall.

The solution $\Psi(x,t)$, when considered for all x , consists of the initial wave packet $\psi(x,t)$ propagating to the right minus an initially unphysical “image” of it $\psi(-x,t)$ propagating to the left. This image becomes the reflected packet as it passes through the wall. For the period of time that this reflected packet overlaps the incident one, the interference between them produces rapid oscillations in the magnitude $|\Psi|$ of the wave function.

Figure 2 shows the result of applying this method to the evolution of a Gaussian wave packet. This was easily calcu-

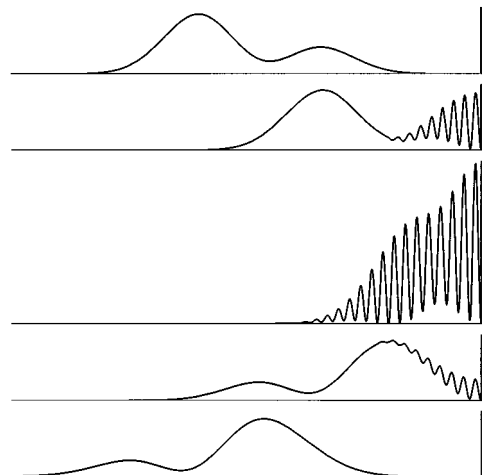


Fig. 3. The evolution of a free, asymmetric wave packet bouncing off a wall, showing that the wave packet is reversed after reflection. The probability density is shown for five equally spaced times. The initial wave packet is the sum of two propagating Gaussians, as in Eq. (2) with $ka = 20$, $b/a = 5/3$, $c = 0.7$.

lated because the evolution of a propagating Gaussian (i.e., a wave function of the form $\exp[ikx - x^2/a^2]$) is simple and is given in many textbooks.² Robinett,¹ in his Fig. 3.4, shows results for a similar case calculated by solving Schrödinger’s equation numerically; he also shows the interesting evolution of the momentum distribution, which is difficult to deal with analytically because it involves the Fourier transform of a truncated propagating Gaussian.

The incident packet will spread as it propagates. More accurately, $(\Delta x)^2 \equiv \langle (x - \langle x \rangle)^2 \rangle$ will be quadratic in time and may decrease for a time but will eventually increase.³ The solution in Eq. (1) shows that the evolution of the shape of the packet is unaffected by its reflection off the wall; the reflected packet $\psi(-x,t)$ continues to evolve in the same way that the incident packet $\psi(x,t)$ would have evolved, except that the packet is spatially reversed. The wave packet used for Fig. 2 took its minimum width at the initial time and will therefore spread as time goes on. The continued spreading after reflection can be seen in the figure.

The simple Gaussian wave packet is symmetric about its mean position and therefore it is not apparent in Fig. 2 that the wave packet is flipped after reflection off the wall. One way to construct an asymmetric wave packet whose evolution is easily calculated is to use a superposition of two Gaussians with the same momentum; thus

$$\psi(x,0) = \exp(ikx) [\exp(-x^2/a^2) + c \exp(-(x-b)^2/a^2)]. \quad (2)$$

The free evolution of two Gaussians does not produce interference fringes in their overlap when they have the same momentum. An example of the evolution of such a wave packet is shown in Fig. 3; the flipping on reflection is now apparent.

The solution in Eq. (1) can also be obtained from the eigenfunctions for the potential step in the limit as the step height is made infinitely large.⁴ For a finite step and energy less than the step in potential, the amplitude of the reflected sinusoidal wave is the same as the incident one, but the phase varies with the energy and therefore the shape of a

wave packet is distorted by the reflection. In the limit of a very high barrier, however, the phase of the reflected sinusoid in the eigenfunction becomes independent of the energy; in fact, the reflected wave is just the opposite of the incoming one so that they cancel at the wall. Then a wave packet is not distorted by the reflection, but has its sign changed. Such an analysis would normally come later in an introductory course.

The image method can be extended to include motion under a force; but the potential must be made symmetric to ensure that $\psi(-x, t)$ satisfies Schrödinger's equation. [Formally, if the parity operator commutes with the Hamiltonian, then the initially odd wave function $\Psi(x, t)$ will remain odd, and therefore remain zero at the origin. For the parity operator to commute with the Hamiltonian, the potential must be even.] A harmonic well (or hill) with a wall at the center is easily handled by this method because the potential is already symmetric. This system is the quantum analogue of a particle attached to the wall by a spring; this was dubbed a "quantum paddle ball" by Robinett.⁵ However, if the given potential $V(x)$ is not symmetric we must construct an "image" potential $V(-x)$ for $x > 0$. An example of this is a particle falling under gravity and bouncing off a rigid table; a suitable potential is $V = mgx$, where x is the height above the table. The method used above applies only if one first solves for the motion of the incident packet with the symmetric potential $V(x) = mg|x|$. It is doubtful whether this approach would be of any use then; one might just as well solve with the wall directly. The difference between the linear and harmonic cases is well illustrated by considering Gaussian incident packets. It is well known that there are Gaussian packets that remain Gaussian in both the harmonic⁶ and the linear⁷ cases. If the wall is now included, then in the harmonic case the reflected packet will remain Gaussian; but

in the case of a falling packet the shape of the reflected packet will not, in general, be Gaussian in shape—it is distorted by the reflection.

More generally, if $\psi(x, t)$ is a wave packet satisfying Schrödinger's equation [with potential $V(x)$] for all x and t , then if the wall is included (at $x=0$), the reflected packet will be just $\psi(-x, t)$ if $V(x)$ is symmetric. But if $V(x)$ is not symmetric, the reflected packet will be the spatial reverse of that solution of Schrödinger's equation [with potential $V(|x|)$] which is initially $\psi(x, 0)$.

This method of images has been used by Kleber⁸ to derive the propagator for a particle in a one-dimensional box. This involves a sum over an infinite number of images, one for each time the packet hits a wall. The method was also used by Greenman⁹ for the evolution of a wave function in a finite well in the no-dispersion approximation. Finally, it should be mentioned that the method can be generalized to higher dimensions. For example, if the free wave packet $\psi(x, y, z, t)$ is incident on a planar wall at $x=0$, its evolution will be $\psi(x, y, z, t) - \psi(-x, y, z, t)$.

¹Richard W. Robinett, *Quantum Mechanics: Classical Results, Modern Systems, and Visualised Examples* (Oxford U.P., New York, 1997), Sec. 3.3.2, pp. 62–64.

²For example, see Ref. 1, Sec. 3.2.2, p. 57.

³Claude Cohen-Tannoudji, Bernard Diu, and Franck Laloë, *Quantum Mechanics* (Wiley, New York, 1977), Vol. I, Complement L_{III}, Exercise 4, p. 342.

⁴Reference 3, Complement H_I, Sec. 2aβ, p. 71.

⁵Reference 1, Sec. 9.3, p. 185.

⁶For example, see Ref. 1, Sec. 10.5, pp. 206–207.

⁷For example, see Ref. 1, Sec. 5.6.2, pp. 100–103.

⁸M. Kleber, "Exact solutions for time-dependent phenomena in quantum mechanics," *Phys. Rep.* **236** (6), 331–393 (1994).

⁹J. V. Greenman, "Non-Dispersive Mirror Wavepackets," *Am. J. Phys.* **40**, 1193–1200 (1972).